GitHub: https://github.com/divarhan123/Tugas-BRSP-Week-4

Judul:

Pancreatic cancer biomarkers in mouse saliva

Pendahuluan

Global gene expression profiling has shown great promise in high-throughput biomarker discovery for early disease detection in body fluids such as saliva, which is accessible, cost-effective, and non-invasive. However, this goal has not been fully realized because saliva, like many clinical samples, contains partially fragmented and degraded RNAs that are difficult to amplify and detect with prevailing technologies. Here, using nanogram scale salivary RNA as a proof-of-principle example, we describe our progress with a novel poly-A tail independent mRNA amplification strategy combined with the Affymetrix GeneChip Exon arrays. We defined a Salivary Exon Core Transcriptome (SECT) with highly similar expression profiles in healthy individuals verified by quantitative PCR. Informatics analysis of SECT provided important mechanistic insight to their potential origin and function. Finally we demonstrated the diagnostic potential of true exon level expression profiling approach with salivary exon biomarkers that accurately discriminated gender in healthy individuals.

Recent studies have demonstrated that discriminatory salivary biomarkers can be readily detected upon the development of systemic diseases such as pancreatic cancer, breast cancer, lung cancer and ovarian cancer. However, the utility of salivary biomarkers for the detection of systemic diseases has been undermined due to the absence of biological and mechanistic rationale why distal diseases from the oral cavity would lead to the development of discriminatory biomarkers in saliva. Here, we examine the hypothesis that pancreatic tumor-derived exosomes are mechanistically involved in the development of pancreatic cancer-discriminatory salivary transcriptomic biomarkers. We first developed a pancreatic cancer mouse model that yielded discriminatory salivary biomarkers by implanting the mouse pancreatic cancer cell line Pan02 into the pancreas of the syngeneic host C57BL/6. The role of pancreatic cancer-derived exosomes in the development of discriminatory salivary biomarkers was then tested by engineered a Pan02 cell line that is suppressed for exosome biogenesis, implanted into the C56BL/6 mouse and examine if the discriminatory salivary biomarker profile was ablated or disrupted. Suppression of exosome biogenesis results in the ablation of discriminatory salivary biomarker development. This study supports that tumor-derived exosomes provide a mechanism in the development of discriminatory biomarkers in saliva and distal systemic diseases.

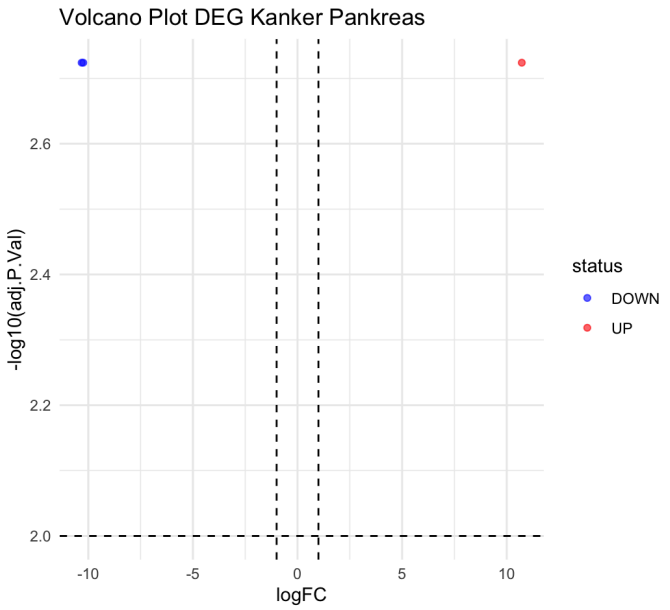
GitHub: https://github.com/divarhan123/Tugas-BRSP-Week-4

Metode

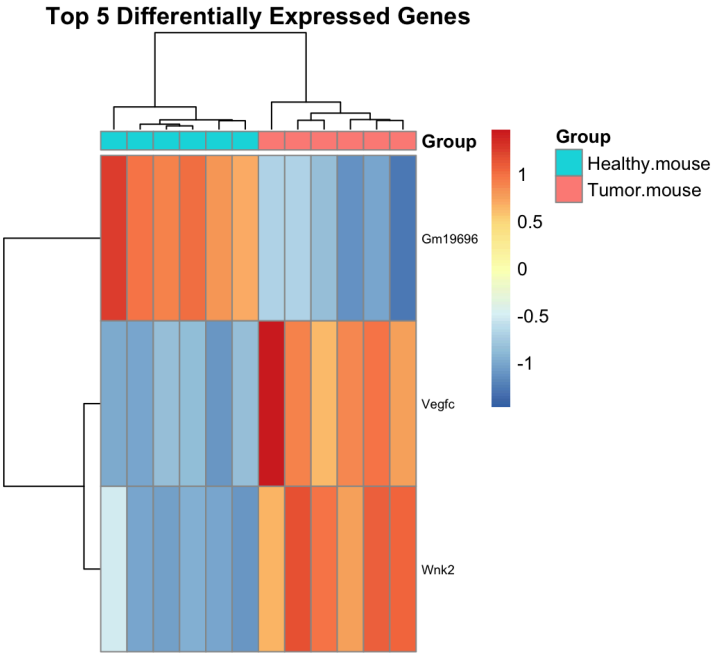
We analyzed saliva from 6 healthy mice and 6 mice with a pancreatic tumor using the Affymetrix Mouse Exon Genome 430 2.0 platform. Array data was processed by dChip. No technical replicates were performed.

Hasil dan Interpretasi

Beberapa gen mengalami *upregulation* dan *downregulation* yang disajikan dalam volcano plot sebagai berikut.

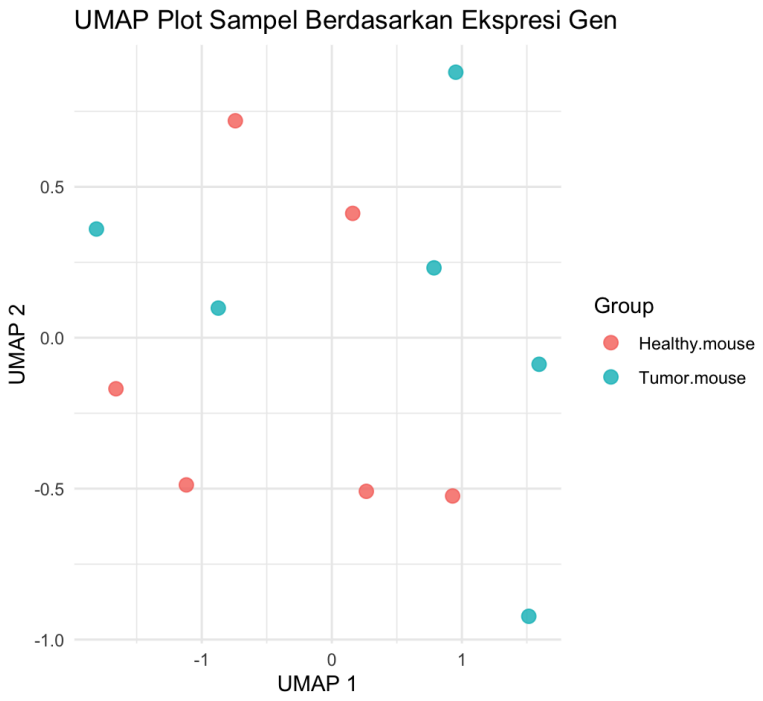


Terdapat 5 *Differentially Expressed Genes* (DEGs) teratas yang ditampilkan dalam bentuk heatmap sebagai berikut.

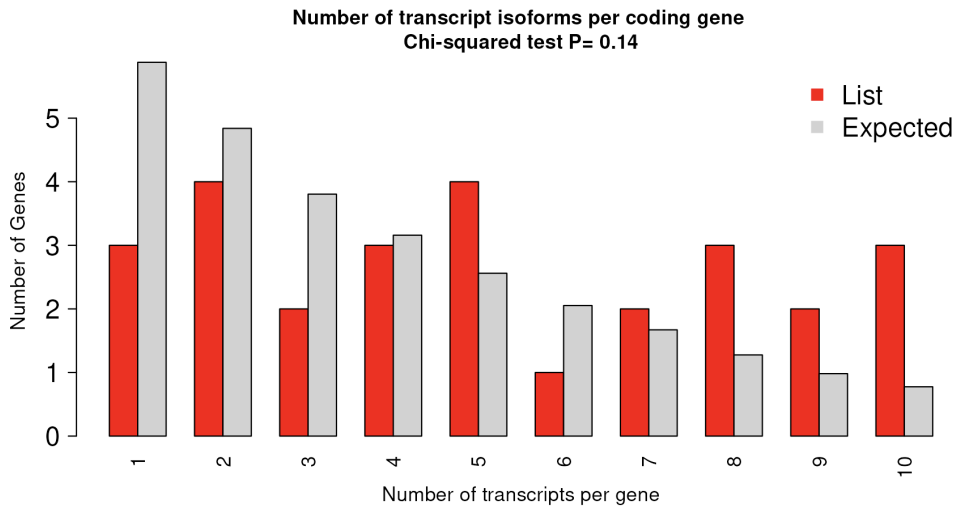


GitHub: https://github.com/divarhan123/Tugas-BRSP-Week-4

Visualisasi UMAP dan distribusi nilai ekspresi gen pada kanker pankreas dan kontrol sehat adalah sebagai berikut.



- Analisis Lanjutan Menggunakan ShinyGO versi 0.85.1



Kesimpulan

Berdasarkan hasil analisis diketahui bahwa gen Vegfc, Gm19696, dan Wnk2 dapat digunakan sebagai biomarker saliva untuk deteksi penyakit kanker pankreas pada tikus.